

LABORATORY OBSERVATION

Course: Full Stack Web Development
Course Code: 23CM4121
Experiment No.: 1
Student Name: P.Dinesh Kumar
Roll No.: A24126552101
Date: 30-08-2026

1. TITLE

Design and Implementation of College Information Portal Using HTML5

2. AIM

To develop a structured, semantic HTML5 web page for the ANITS College Information Portal that incorporates text formatting, department lists, student detail tables, interactive registration forms, and multimedia audio/video players.

3. OBJECTIVE

The objectives of this experiment are:

- To understand core HTML5 semantic tags and document layout structure.
- To implement formatted text, mathematical/chemical formulas using sup and sub tags.
- To create structured tabular data using table, tr, th, and td elements.
- To design an interactive student registration form with various HTML5 input types.
- To embed and control multimedia audio and video players.

4. THEORY

HTML5 is the standard markup language for creating modern web pages. It introduces semantic elements such as `<header>`, `<nav>`, `<section>`, `<article>`, and `<footer>` to structure content meaningfully.

Forms in HTML5 provide rich input controls including text, email, password, radio buttons, checkboxes, dropdown select, color pickers, and date pickers for client-side data capture.

HTML5 also provides native `<audio>` and `<video>` elements with browser controls for seamless media playback without third-party plugins.

Application Flow:

Browser -> HTML5 Document -> Semantic Elements -> Tables & Forms -> Media Players -> Rendered Webpage

5. ALGORITHM / PROCEDURE

1. Create an HTML5 document and configure the title as 'ANITS College Information Portal'.
2. Add a header section with institution title and introductory description.
3. Create a navigation bar with hyperlink anchors to sections and external portals.
4. Format about text using bold, italic, underline, superscript, and subscript tags.
5. Create ordered list for departments and unordered list for campus facilities.
6. Construct a student details table with borders, column headers, and records.
7. Build a comprehensive student registration form with text, email, password, branch select, date, color, and range inputs.
8. Embed audio and video tags with controls attribute.
9. Open in web browser and verify all elements and form controls.

6. PROGRAM

task1.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <title>ANITS College Information Portal</title>
</head>

<body>

<header>

  <h1>Anil Neerukonda Institute of Technology and Sciences (ANITS)</h1>

  <p>Welcome to the ANITS College Information Portal.</p>

  <nav>
    <a href="#about">About</a> |
    <a href="#courses">Courses</a> |
    <a href="#students">Students</a> |
    <a href="#register">Registration</a> |
    <a href="#media">Media</a>
  </nav>

</header>

<hr>

<section id="about">

  <article>

    <h2>About ANITS</h2>

    <p>
      <b>ANITS</b> is one of the well-known engineering colleges in Andhra Pradesh.
      It provides <i>quality education</i> with a <u>good learning environment</u>
      for students.
    </p>

  </article>

</section>
```

```
</article>

</section>

<hr>

<section id="courses">

    <h2>Courses and Facilities</h2>

    <h3>College Facilities</h3>

    <ul>
        <li>Central Library</li>
        <li>Computer Laboratories</li>
        <li>Sports Ground</li>
    </ul>

    <h3>Popular Branches</h3>

    <ol>
        <li>Computer Science and Engineering</li>
        <li>Electronics and Communication Engineering</li>
        <li>Electrical and Electronics Engineering</li>
    </ol>

    <p>
        Visit
        <a href="https://www.anits.edu.in" target="_blank">ANITS Official Website</a>
        or
        <a href="https://www.google.com" target="_blank">Google</a>.
    </p>

</section>

<hr>

<section id="students">

    <h2>Student Details</h2>

    <table border="1" cellpadding="8" cellspacing="0">

        <tr>
            <th>Roll No</th>
            <th>Name</th>
            <th>Branch</th>
            <th>Email</th>
        </tr>

        <tr>
            <td>101</td>
            <td>Dinesh Kumar</td>
            <td>CSE</td>
            <td>dinesh@gmail.com</td>
        </tr>

        <tr>
            <td>102</td>
            <td>Rohit</td>
            <td>ECE</td>
```

```

        <td>rohit@gmail.com</td>
    </tr>

    <tr>
        <td>103</td>
        <td>Charan</td>
        <td>EEE</td>
        <td>charan@gmail.com</td>
    </tr>

    <tr>
        <td>104</td>
        <td>Vinay</td>
        <td>IT</td>
        <td>vinay@gmail.com</td>
    </tr>

</table>

<br>





</section>

<hr>

<section id="register">

    <h2>Student Registration Form</h2>

    <form>

        <label>Name:</label><br>
        <input type="text"><br><br>

        <label>Email:</label><br>
        <input type="email"><br><br>

        <label>Password:</label><br>
        <input type="password"><br><br>

        <label>Gender:</label><br>
        <input type="radio" name="gender"> Male
        <input type="radio" name="gender"> Female
        <br><br>

        <label>Hobbies:</label><br>
        <input type="checkbox"> Reading
        <input type="checkbox"> Music
        <input type="checkbox"> Sports
        <br><br>

        <label>Branch:</label><br>

        <select>
            <option>CSE</option>
            <option>ECE</option>
            <option>EEE</option>

```

```
        <option>IT</option>
    </select>

    <br><br>

    <label>Date of Birth:</label><br>
    <input type="date"><br><br>

    <label>Age:</label><br>
    <input type="number"><br><br>

    <label>Favorite Color:</label><br>
    <input type="color"><br><br>

    <label>Programming Skill:</label><br>
    <input type="range" min="1" max="10"><br><br>

    <label>Comments:</label><br>
    <textarea rows="4" cols="40"></textarea>

    <br><br>

    <input type="submit" value="Submit">

    <input type="reset" value="Reset">

</form>

</section>

<hr>

<section id="media">

    <h2>College Media</h2>

    <p>College Welcome Audio</p>

    <audio controls>
        <source src="welcome.mp3" type="audio/mpeg">
    </audio>

    <br><br>

    <p>Campus Introduction Video</p>

    <video width="400" controls>
        <source src="campus.mp4" type="video/mp4">
    </video>

</section>

<hr>

<footer>

    <p>Designed by Dinesh Kumar</p>

    <p>B.Tech Student | ANITS</p>

</footer>
```

```
</body>

</html>
```

7. CODE EXPLANATION

Code / Syntax	Explanation
<header>, <nav>, <footer>	Semantic HTML5 layout container elements.
<sup>, <sub>	Displays text as superscript (X ²) or subscript (H ₂ O).
, ,	Creates ordered (numbered) and unordered (bulleted) lists.
<table>, <tr>, <th>, <td>	Structures tabular data into rows, header cells, and standard data cells.
<input type='radio' name='gender'>	Radio button group allowing only one gender selection.
<select>, <option>	Drop-down select menu for branch selection.
<audio controls>, <video controls>	Native media playback controls for audio and video files.

8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Page Load	Open task1.html in browser	Header, nav, and sections render cleanly	Pass
TC-02	Lists & Formulas	Verify lists and formulas	Ordered/unordered lists and formatting display properly	Pass
TC-03	Table Rendering	Check student details table	Table renders 4 columns with row borders	Pass
TC-04	Form Inputs	Fill form & click Submit	Inputs accept data, radio selection works	Pass
TC-05	Media Playback	Click play on audio/video	Media plays with default browser controls	Pass

9. OUTPUT

Anil Neerukonda Institute of Technology and Sciences (ANITS)

Welcome to the ANITS College Information Portal.

About | Courses | Students | Registration | Media

About ANITS

ANITS is one of the well-known colleges in Andhra Pradesh. It provides quality education with a good learning environment for students.

Courses and Facilities

College Facilities

- Central Library
- Computer Laboratories
- Sports Ground

Popular Branches

- Computer Science and Engineering
- Electronics and Communication Engineering
- Electrical and Electronics Engineering

Visit [ANITS Official Website](#) or [Google](#).

Student Details

Roll No	Name	Branch	Email
101	Dinesh Kumar	CSE	dinesh@gmail.com
102	Rohit	ECE	rohit@gmail.com
103	Charan	EEE	charan@gmail.com
104	Vinay	IT	vinay@gmail.com

ANITS Campus

Student Registration Form

Name:

Email:

Password:

Gender:

☐ Male ☐ Female

Hobbies:

☐ Reading ☐ Music ☐ Sports

Branch:

CSE ▾

Date of Birth:

mm / dd / yyyy ▾

Age:

Favorite Color:

Programming Skill:

Comments:

Submit

Reset

College Media

College Welcome Audio

> 0:00 / 0:00 🔊 ⋮

Campus Introduction Video

> 0:00 🔊 🔄 ⋮

Designed by Dinesh Kumar

B.Tech Student | ANITS

Output 1: College Information Portal Webpage & Registration Form

10. RESULT

Thus, the College Information Portal was successfully implemented using HTML5, incorporating text formatting, lists, tables, registration form inputs, and multimedia elements, and verified across all test cases.